



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

SPRING BOOT PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A covering auto-config, JPA, security, and testing

TOTAL: 10 QUESTIONS

Q1 What is Spring Boot and how does it differ from the Spring Framework?

Threat Model

Q6 How do you configure a relational database in Spring Boot?

Observability

Q2 How does Spring Boot auto-configuration work under the hood?

Architecture

Q7 How do you replace the default embedded Tomcat with Jetty?

Lifecycle

Q3 What does `@SpringBootApplication` combine and why is each part needed?

Configuration

Q8 How do you implement global REST exception handling in Spring Boot?

Compliance

Q4 How does Spring Boot externalize and prioritize configuration?

Hardening

Q9 What are Spring Boot profiles and how do you activate one?

Testing

Q5 What is Spring Boot Actuator and what does it expose?

SSO / IdP

Q10 How do you write a Spring Boot integration test?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Spring Boot is a Spring module

Mitigation

Spring Boot is a Spring module that provides auto-configuration, embedded servers, and starter POMs. Unlike plain Spring, it needs zero XML and runs as a self-contained JAR, removing the need to deploy to a separate app server.

A6 Add spring-boot-starter-data-jpa and a JDBC

Observability

Add spring-boot-starter-data-jpa and a JDBC driver. Set spring.datasource.url, username, and password in application.properties. Boot auto-wires a DataSource, EntityManagerFactory, and JpaTransactionManager without any explicit @Bean definitions.

A2 @EnableAutoConfiguration loads candidate

Primitives

@EnableAutoConfiguration loads candidate classes from AutoConfiguration.imports. Each class uses @ConditionalOnClass or @ConditionalOnMissingBean to activate only when specific dependencies or beans are present on the classpath.

A7 Exclude spring-boot-starter-tomcat from the web

Automation

Exclude spring-boot-starter-tomcat from the web starter, then add spring-boot-starter-jetty. Auto-configuration detects Jetty on the classpath and selects it. Undertow is a third option available the same way.

A3 It merges @Configuration (declares beans),

Hardening

It merges @Configuration (declares beans), @EnableAutoConfiguration (activates Boot's auto-config machinery), and @ComponentScan (recursively discovers @Component, @Service, @Repository in the package tree).

A8 @RestControllerAdvice marks a class as a

Governance

@RestControllerAdvice marks a class as a cross-cutting error handler. Methods annotated @ExceptionHandler(SomeEx.class) intercept matched exceptions and return a ResponseEntity with a structured error body and HTTP status code.

A4 Boot loads sources in ascending-priority order:

Gotchas

Boot loads sources in ascending-priority order: application.yml, profile files, OS environment variables, and CLI args each override the previous. @ConfigurationProperties binds a prefix to a typed POJO with IDE-validation support.

A9 Profiles separate environment configuration.

Assessment

Profiles separate environment configuration. application-prod.yml is only loaded when the prod profile is active. Activate via spring.profiles.active=prod in properties, the SPRING_PROFILES_ACTIVE env var, or --spring.profiles.active=prod CLI arg.

A5 Actuator adds management HTTP endpoints:

Federation

Actuator adds management HTTP endpoints: /actuator/health (liveness/readiness), /actuator/metrics (counters, timers), /actuator/loggers (change log levels at runtime), and /actuator/env. Each endpoint is securable and toggleable.

A10 @SpringBootTest starts a full application

Optimization

@SpringBootTest starts a full application context. Add @AutoConfigureMockMvc to inject a MockMvc for HTTP-level assertions without a real server. Use @MockBean to replace infrastructure beans (e.g., database, email) with Mockito doubles.